

§9.3 Analysis of variance (ANOVA) tests

One-factor tests (1 factor influences performance of the experiment, e.g. different doses of a drug / different age groups / etc)

We want to know whether this difference results in different means

<u>Data:</u>		Sample means	True means
1st experiment:	$X_{11} \ X_{12} \dots \ X_{1n_1}$	$\bar{X}_1$	μ_1
2nd:	$X_{21} \ X_{22} \dots \ X_{2n_2}$	$\bar{X}_2$	μ_2
$\vdots$			
m-th:	$X_{m1} \ X_{m2} \dots \ X_{mn_m}$	$\bar{X}_m$	μ_m

Grand sample mean: $\frac{1}{n} \sum_{i=1}^m \sum_{j=1}^{n_i} X_{ij} =: \bar{X}$

$n = n_1 + \dots + n_m$ (total number of samples)

Assume that $X_{ij} \sim N(\mu_i, \sigma^2)$

Hypothesis: $H_0: \mu_1 = \mu_2 = \dots = \mu_m = \mu$

Construct test statistic:

$SS(TO) := \sum_{i=1}^m \sum_{j=1}^{n_i} (X_{ij} - \bar{X})^2$ total sum of squares

$\underbrace{\quad}_{-\bar{X}_i + \bar{X}_i}$

$$SS(T) = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2 + \sum_i \sum_j (\bar{x}_i - \bar{x})^2 + \text{cross terms}$$

$\parallel$ $\parallel$
 $SS(E)$ $SS(T)$
 "within treatments sum of squares" "between treatments sum of squares"
 or "error sum of squares"

(see p 436 computations)

3 facts about $SS(\cdot)$: recall §5.5

① if H_0 , $\frac{SS(T)}{\sigma^2} \sim \chi^2(n-1)$ none big sample from $N(\mu, \sigma^2)$

② in any case, $\frac{SS(E)}{\sigma^2} \sim \chi^2(n-m)$
 $\sim \sum_{i=1}^m \chi^2(n_i-1)$

③ if H_0 , $\frac{SS(T)}{\sigma^2} \sim \chi^2(m-1)$

independent (Theorem 9.3-1)

Test statistic is $F = \frac{SS(T)/(m-1)}{SS(E)/(n-m)}$

if H_0 ,
 $\frac{SS(T)/\sigma^2}{m-1} \cdot \frac{1}{\frac{SS(E)/\sigma^2}{n-m}}$

Critical region?

Observe that, given H_0 , both

$\frac{SS(T)}{m-1}$ and $\frac{SS(E)}{n-m}$ are estimators for same σ^2

unbiased, $E\left(\frac{SS(T)}{m-1}\right) = \sigma^2 = E\left(\frac{SS(E)}{n-m}\right)$

$\sim \frac{\chi^2(m-1)/(m-1)}{\chi^2(n-m)/(n-m)} \sim F_{(m-1, n-m)}$

F-distribution (Table VII)

So, if H_0 , $F = \frac{MS(T)}{MS(E)} = \frac{SS(T)/(m-1)}{SS(E)/(n-m)} \sim \frac{\sigma^2}{\sigma^2} = 1$

↑
mean square error

Critical region: what goes different when H_0 is not satisfied?

$$E(SS(T)) = E\left(\sum_{i=1}^m n_i (\bar{X}_i - \bar{X})^2\right) = \quad (\text{see p. 438})$$

$$= E\left(\sum_{i=1}^m n_i X_i^2 - n_i \bar{X}^2\right) = (m-1)\sigma^2 + \underbrace{\sum_{i=1}^m n_i (\mu_i - E\bar{X})^2}_{\sigma_0^2}$$

$$E\left(\frac{SS(T)}{m-1}\right) > \sigma^2$$

$$E\left(\frac{SS(E)}{n-m}\right) = \sigma^2 \text{ always}$$

→ so if not H_0 ,
 F tends to be large

Critical region: $\boxed{\{F \geq F_{\alpha}(m-1, n-m)\}}$

Example

(9.3-1)

$$\alpha = 0.05$$

$X_1:$	5	9	6	8
$X_2:$	11	13	10	12
$X_3:$	10	6	9	9

$H_0: \mu_1 = \mu_2 = \mu_3$
against all possible
alternative hypotheses

$$m=3, n=4 \cdot 3 = 12$$

$$SS(T) = \sum_{i=1}^m \sum_{j=1}^{n_i} X_{ij}^2 - \frac{1}{n} \left(\sum \sum X_{ij} \right)^2$$

sum everything!

$$SS(T) = \sum \frac{1}{n_i} \left(\sum_{j=1}^{n_i} X_{ij} \right)^2 - \frac{1}{n} \left(\sum \sum X_{ij} \right)^2$$

$$SS(E) = SS(TO) - SS(T)$$

note that
we always
normalize
by the number
of terms!

$$\sum \sum x_{ij}^2 = 1038 (= 25 + 81 + 36 + \dots)$$

$$(\sum \sum x_{ij})^2 = 108^2$$

$$\frac{108^2}{12} = \cancel{2976} 972$$

$$SS(TO) = \underline{\underline{66}}$$

$$\left. \begin{aligned} \left(\sum_{j=1}^{n_1} x_{ij} \right)^2 &= 28^2 \\ \left(\sum_{j=1}^{n_2} x_{ij} \right)^2 &= 46^2 \\ \left(\sum_{j=1}^{n_3} x_{ij} \right)^2 &= (34)^2 \end{aligned} \right\}$$

$$SS(T) = \frac{784}{4} + \frac{2116}{4} + \frac{1156}{4} = 972$$

$$= 1014 - 972 = \underline{\underline{42}}$$

$$SS(E) = \underline{\underline{24}}$$

$$F = \frac{42/2}{24/8} = \frac{21.9}{3} = 7.8$$

$$F_{0.05}(2, 8) = 4.46$$

$F > F_{0.05} \Rightarrow$
reject H_0 .

ANOVA table

	Degrees of freedom	MS	F
SS(T)	$n - 1$	$\frac{SS(T)}{n - 1}$	$\frac{MS(T)}{MS(E)}$
SS(E)	$n - m$	$\frac{SS(E)}{n - m}$	—
SS(TO)	$n - 1$	—	—

SS(V)	3	21	7.8
SS(E)	9	24/9	
SS(TO)	11		